**RELATED RATES QUIZ**

BLAKE FARMAN

*Lafayette College*

Name: Solutions

**1.** The volume of a tree is given by $V = \frac{1}{12\pi}C^2h$, where $C$ is the circumference of the tree in meters at ground level, and $h$ is the height of the tree in meters. Both $C$ and $h$ are functions of time, $t$, in years.

(a) Find a formula for $\frac{\mathrm{d}V}{\mathrm{d}t}$. What does it represent in practical terms?

$$\frac{dV}{dt} = \frac{d}{dt}\left(\frac{1}{12\pi}C^2h\right) = \frac{1}{12\pi}\frac{d}{dt}\left(C^2h\right) = \frac{1}{12\pi}\left(\frac{d}{dt}(C^2)h + C^2\frac{d}{dt}(h)\right)$$

$$= \frac{1}{12\pi}\left(2Ch\frac{dC}{dt} + C^2\frac{dh}{dt}\right) = \boxed{\frac{Ch}{6\pi}\frac{dC}{dt} + \frac{C^2}{12\pi}\frac{dh}{dt}.}$$

This represents the rate of change of the tree's volume w.r.t. time.

(b) Suppose the circumerence grows at a rate of 1/5 meter per year, and the height grows at a rate of 4 meters per year. How fast is the volume of the tree growing when the circumference is 5 meters and the height is 22 meters?

$$\frac{1}{12\pi}\left(2(5)(22)\left(\frac{1}{5}\right) + (5)^2(4)\right) = \frac{1}{12\pi}(44+100) = \frac{144}{12\pi} = \frac{12^2}{12\pi} = \boxed{\frac{12}{\pi}}$$